Classification of Primitive Pythagorean Triples

An End-to-End Meta-Math Walkthrough

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Abstract

Parity creates integral half-sums. Coprimality lets a square product split into two squares; recovering the parameters and proving the converse completes the classification. This report presents a complete proof relative to explicit imports, a dependency graph, versioned node contracts, five-component activities, exact examples, and an illustrative repair. The interactions are authored explanations, not a transcript of deployed agents or a proof-assistant certificate.

Project NT-05 Report version 1 Number Theory

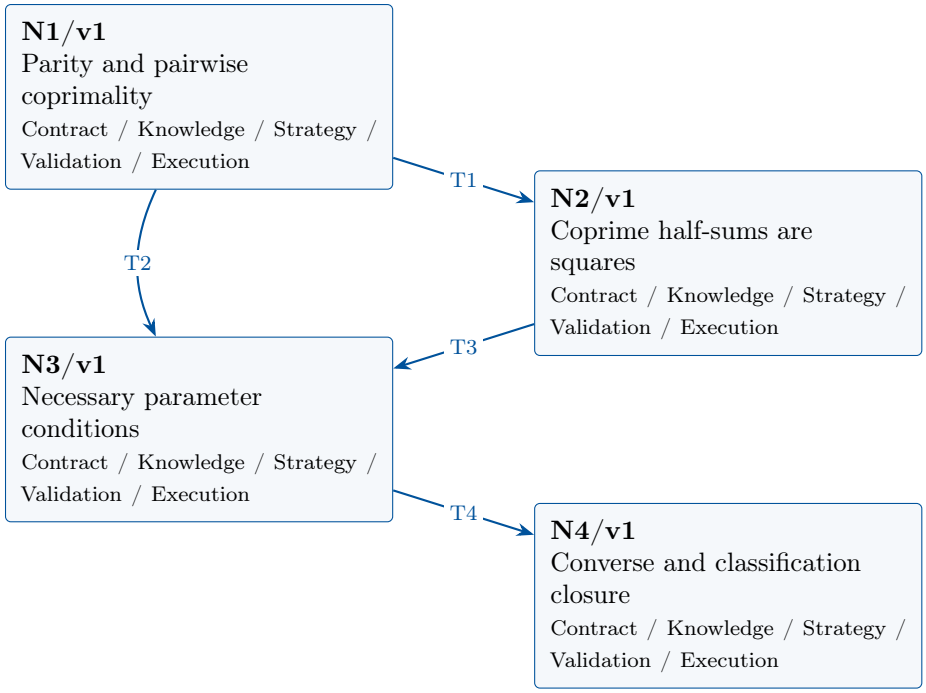
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *Let a, b, c be positive integers with $a^2 + b^2 = c^2$ and $\gcd(a, b, c) = 1$. After exchanging the legs if necessary, there exist integers $m > n \geq 1$ with $\gcd(m, n) = 1$ and opposite parity such that $a = m^2 - n^2$, $b = 2mn$, $c = m^2 + n^2$. Conversely, every such parameter pair gives a primitive positive triple.*

A. Foundations and permitted imports

1. NT-02/N1–N3, v1: prime factorization exists and is unique, and Euclid's lemma holds. In particular $p \mid x^2$ implies $p \mid x$; in a square, every prime exponent is even.
2. An integer square is 0 or 1 modulo 4. A primitive triple has no common prime divisor. Coprimality arguments use the divisibility and gcd conventions of NT-01.

II. NODE INTERACTION WORKFLOW



Blue arrows are accepted mathematical dependencies, with exact transfers T1–T4 below. Repair messages are described separately; they are not premises of the proof.

A. Versioned evidence-transfer ledger

T1: N1/E1 v1 to N2.1–2
Claim. After exchanging legs, b is even, a, c are odd, and every pair among a, b, c is coprime.
Compatibility. Odd half-sums are integral and the required pairwise gcd is one.
Evidence. Node N1 activities below and the continuous proof in Section IV.

T2: N1/E1 v1 to N3.3
Claim. After exchanging legs, b is even, a, c are odd, and every pair among a, b, c is coprime.
Compatibility. The recovered odd leg and hypotenuse constrain parameter parity.
Evidence. Node N1 activities below and the continuous proof in Section IV.

T3: N2/E2 v1 to N3.1–2
Claim. For $u = (c + a)/2$, $v = (c - a)/2$, one has $u = m^2$, $v = n^2$ for coprime integers $m > n \geq 1$.
Compatibility. Use positive integer square roots with their order and gcd.
Evidence. Node N2 activities below and the continuous proof in Section IV.

T4: N3/E3 v1 to N4.3
Claim. Every primitive triple has $a = m^2 - n^2$, $b = 2mn$, $c = m^2 + n^2$, with $m > n \geq 1$, coprime and of opposite parity.
Compatibility. N3 supplies only the forward half; N4 proves the converse independently before joining.
Evidence. Node N3 activities below and the continuous proof in Section IV.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Parity and pairwise coprimality

Object and role. Choose a consistent leg orientation and prove the arithmetic constraints needed for half-sums.

Inputs. A positive primitive triple $a^2 + b^2 = c^2$; prime-divisor facts.

C — Contract

Publish the orientation and pairwise gcd facts, not just collective primitiveness.

K — Knowledge

Retain square residues modulo 4 and $p \mid x^2 \Rightarrow p \mid x$.

S — Strategy

Exclude both-even and both-odd legs; transfer a shared prime through the equation.

V — Validation

Renaming the legs is allowed; a common divisor must be ruled out for each pair.

E — Execution

Both even legs would make c even, contradicting primitiveness. Continue with the numbered deductions below, retaining the exact export.

Numbered mathematical activities

N1.1 Both even legs would make c even, contradicting primitiveness.

N1.2 Both odd legs give $c^2 \equiv 2 \pmod 4$, impossible; rename the even leg b.

N1.3 Then a, c are odd. A prime dividing two entries divides the square, and hence the value, of the third; all pairs are coprime.

Export E1/v1. After exchanging legs, b is even, a, c are odd, and every pair among a, b, c is coprime.

Acceptance gate. Parity and all three pairwise gcd conclusions follow from the original primitive-triple hypothesis.

Targeted repair. If the input is not primitive, stop this classification or explicitly divide out the common factor under a new task; recheck N2 and N3.

B. N2/v1: Coprime half-sums are squares

Object and role. Extract integer parameters from a square product with disjoint prime support.

Inputs. N1/E1 and NT-02 unique prime factorization.

C — Contract

Publish integer square parameters only after positivity, integrality, and coprimality are proved.

K — Knowledge

Retain a, c odd, $c > a > 0$, and $\gcd(a, c) = 1$.

S — Strategy

Use half-sums to produce coprime factors whose product is a square.

V — Validation

A square product alone is insufficient: $2 \cdot 8 = 16$ with neither factor square.

E — Execution

Set $u = (c + a)/2$, $v = (c - a)/2$. Then $u > v > 0$ are integers and $uv = (b/2)^2$. Continue with the numbered deductions below, retaining the exact export.

Numbered mathematical activities

N2.1 Set $u = (c + a)/2$, $v = (c - a)/2$. Then $u > v > 0$ are integers and $uv = (b/2)^2$.

N2.2 A positive common divisor of u, v divides $u + v = c$ and $u - v = a$, so it is 1.

N2.3 Unique factorization gives even prime exponents in each disjoint factor; take positive square roots $u = m^2$, $v = n^2$.

Export E2/v1. For $u = (c + a)/2$, $v = (c - a)/2$, one has $u = m^2$, $v = n^2$ for coprime integers $m > n \geq 1$.

Acceptance gate. The square roots are positive integers and $\gcd(m, n) = 1$, since any common prime would divide u, v.

Targeted repair. Restore the coprimality proof before splitting the square product; invalidate N3 and the classification assembly in N4.

C. N3/v1: Necessary parameter conditions

Object and role. Recover all three entries and enforce the parameter parity restriction.

Inputs. N1/E1 and N2/E2.

C — Contract

Publish the full parameter contract, including parity and the chosen leg orientation.

K — Knowledge

Retain the half-sum equations and the positivity of b, m, n.

S — Strategy

Recombine sums and differences and choose the positive square root for the even leg.

V — Validation

The polynomial identity alone does not imply primitiveness or opposite parity.

E — Execution

From $u = m^2, v = n^2$, obtain $a = u - v$ and $c = u + v$. Continue with the numbered deductions below, retaining the exact export.

Numbered mathematical activities

N3.1 From $u = m^2, v = n^2$, obtain $a = u - v$ and $c = u + v$.

N3.2 Because $(b/2)^2 = m^2 n^2$ and all terms are positive, $b = 2mn$.

N3.3 Both parameters even contradict coprimality; both odd would make a, c even. They have opposite parity.

Export E3/v1. Every primitive triple has $a = m^2 - n^2$, $b = 2mn$, $c = m^2 + n^2$, with $m > n \geq 1$, coprime and of opposite parity.

Acceptance gate. All displayed formulas and every parameter condition are derived, without guessing a square-root sign.

Targeted repair. If a parameter condition is missing, withhold the forward classification and recheck its use in N4.

D. N4/v1: Converse and classification closure

Object and role. Prove that admissible parameters generate primitive triples, then join both directions.

Inputs. Fresh parameters $m > n \geq 1$, coprime and opposite parity; N3/E3 for the forward direction.

C — Contract

Publish a classification only after both the forward construction and converse are established.

K — Knowledge

Retain Euclid's lemma, opposite parity, and the parameter gcd; keep the converse independent of assuming a primitive output.

S — Strategy

Expand the identity and rule out common prime divisors of the two constructed legs.

V — Validation

An identity check alone misses nonprimitive triples; treat the prime 2 separately.

E — Execution

Expand $(m^2 - n^2)^2 + (2mn)^2 = (m^2 + n^2)^2$; positivity follows from $m > n \geq 1$. Continue with the numbered deductions below, retaining the exact export.

Numbered mathematical activities

N4.1 Expand $(m^2 - n^2)^2 + (2mn)^2 = (m^2 + n^2)^2$; positivity follows from $m > n \geq 1$.

N4.2 An odd prime dividing both legs divides m or n, and then the other, contradicting coprimality.

N4.3 The leg $m^2 - n^2$ is odd, excluding the common prime 2. Thus the legs, and hence the triple, are primitive; join N3's forward direction.

Export E4/v1. Exactly the primitive positive triples, up to exchanging legs, arise from the stated formulas.

Acceptance gate. Identity, positivity, and absence of every common prime are proved for arbitrary admissible parameters.

Targeted repair. If both parameters are odd, reject the input under this contract; do not publish a nonprimitive example as a primitive triple.

IV. CONTINUOUS MATHEMATICAL PROOF

Proof. Let (a, b, c) be a positive primitive triple. If both legs were even, the equation would make c even. If both were odd, their squared sum would be 2 modulo 4, impossible for a square. Hence exactly one leg is even; call it b. Then a, c are odd. If a prime divided two entries, the equation would make it divide the square of the third, and Euclid's lemma would make it divide the third entry too. This contradicts primitiveness. Thus all pairs are coprime (N1).

Since $c^2 = a^2 + b^2 > a^2$, positivity gives $c > a$. Define

$$u = \frac{c + a}{2}, \qquad v = \frac{c - a}{2}.$$

These are integers with $u > v > 0$ and $uv = (b/2)^2$. A positive common divisor of u, v divides $c = u + v$ and $a = u - v$, so $\gcd(u, v) = 1$. Every prime occurs in at most one of their factorizations. Since the product is a square, uniqueness of factorization forces each exponent in each factor to be even, including the empty factorization if a factor is 1. Therefore $u = m^2$, $v = n^2$ for positive integers $m > n$. A common prime of m, n would divide u, v, so $\gcd(m, n) = 1$ (N2).

It follows that $a = m^2 - n^2$, $c = m^2 + n^2$, and, taking the positive square root of $(b/2)^2 = m^2n^2$, $b = 2mn$. The parameters cannot both be even by coprimality, or both odd because a, c are odd. Thus they have opposite parity (N3).

Conversely, take any parameters meeting these conditions (N4). The formulas give positive integers, and direct expansion gives the Pythagorean equation. If an odd prime p divided both $m^2 - n^2$ and $2mn$, Euclid's lemma would give $p \mid m$ or $p \mid n$. Either case, together with $p \mid m^2 - n^2$, implies that p divides the square and then the value of the other parameter, contradicting their gcd. The prime 2 cannot divide both legs, because opposite parity makes $m^2 - n^2$ odd. Any common divisor greater than 1 would have a prime divisor, so the two legs are coprime and the triple is primitive. Both directions prove the classification. The parameters are also recoverable uniquely after choosing the even leg, as $m = \sqrt{(c + a)/2}$ and $n = \sqrt{(c - a)/2}$.

□

V. WORKED ORCHESTRATION

Start with $(a, b, c) = (5, 12, 13)$. N1 checks the even leg and pairwise gcds. N2 obtains

$$u = \frac{13 + 5}{2} = 9, \qquad v = \frac{13 - 5}{2} = 4,$$

so $m = 3, n = 2$. N3 recovers $5 = 9 - 4$, $12 = 2 \cdot 3 \cdot 2$, and $13 = 9 + 4$. The parameters are coprime and of opposite parity.

In the converse direction, $m = 4, n = 1$ produces $(15, 8, 17)$, and $15^2 + 8^2 = 225 + 64 = 289 = 17^2$. The parameter test and N4's prime-divisor argument justify primitiveness, beyond the numerical identity.

A. Boundary and hypothesis audit

- $m = 2, n = 1$ gives $(3, 4, 5)$ and includes the allowed boundary $n = 1$. The case $n = 0$ would give a zero leg and is excluded.
- $m = 3, n = 1$ gives $(8, 6, 10)$, showing that coprime parameters with the same parity need not give a primitive triple.
- $m = 6, n = 3$ gives $(27, 36, 45)$: opposite parity alone does not imply coprimality. Both conditions are enforced. A square product without coprimality, such as $2 \cdot 8 = 16$, cannot justify square factors.

VI. ILLUSTRATIVE FAILURE AND REPAIR

This is a deliberately authored teaching trace. A diagnostic asks for correction; it cannot prove a predecessor by importing its descendant's conclusion. A revised export must be checked before its consumers are reused.

Candidate. N2 infers that u, v are squares from uv being a square, omitting the gcd check.

Detection. Validation supplies $u = 2, v = 8$, for which $uv = 16$ but neither factor is square.

Correction. Knowledge retrieves $\gcd(a, c) = 1$ from N1. Execution proves $\gcd(u, v) = 1$ via their sum and difference; Strategy then applies disjoint prime-exponent parity.

Recheck. Re-establish N2's integer parameters and N3's formulas and parity. Recheck the N4 forward/converse assembly.

Publication. Contract publishes square splitting only for coprime positive factors. This fills the missing inference under the original primitive-triple assumptions.

VII. REUSABLE RESULT AND PROVENANCE

Reusable exports retain each node's version 1 statement, assumptions, and proof. The final export is the theorem in Section 1, proved relative to its declared imports.

Source record. Adapted from the original project Markdown:
`classification-of-primitive-pythagorean-triples.md`

Typography, workflow conventions, and component cards follow the workspace Meta-Math manuscript. This authored exposition does not claim formal certification.